

Required Syllabus Statements

Please check the campus-wide required syllabus statements at <https://www.colorado.edu/academicaffairs/about/policies-customs-guidelines/required-syllabus-statements>. If you require special accommodations due to medical conditions, religious observances, etc., please contact the instructor in advance.

Honor Code and Student Code of Conduct

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the [Honor Code and Student Code of Conduct](#). The Honor Code and the Student Code of Conduct both lay out the fair, respectful and ethical behaviors that allow the Buff community to thrive and encourage each CU Boulder student to succeed.

AI-specific Note

AI offers both benefits and challenges for learning. When used appropriately, it can significantly enhance the educational experience, but its misuse or overreliance can hinder crucial aspects of learning and development, so you should use good judgment to maximize your learning benefit. AI use will be specified on a per-assignment basis, and AI tools may be incorporated into in-class activities.

Class Syllabus

Course Name and Number: ECEN 4/5632 Digital Signal Processing and Filtering – Fall 2026

Credit and Hours: 3 Credits, Class TTh 9:30–10:45 am **ECCR 131/online**.

Instructor: Prof. Peter Mathys, ECOT 438, mathys@colorado.edu

Office Hours: TTh 12-2 pm (tentative)

Textbook: Oppenheim/Schafer, “Discrete-Time Signal Processing,” 3rd ed., 2010, ISBN-13: 987-0-13-198842-2.

Supplementary References: Gold/Morgan/Ellis, “Speech and Audio Signal Processing,” 2nd ed., 2011, ISBN-13: 978-0-470-19536-9

Raschka/Liu/Mirjalili, “Machine Learning with PyTorch and SciKit-Learn,” 2022, ISBN-13: 978-1-80181-931-2

TA: Peter Braza (Peter.Braza@colorado.edu)

Computer Usage: Python

Prerequisite: ECEN 3300, Linear Systems

Course Description: This course introduces digital signal processing and filtering fundamentals with applications to speech and music signals, building toward modern machine learning approaches. The course emphasizes understanding core DSP concepts through mathematical analysis while providing hands-on experience with audio processing and Machine Learning applications. Restricted to College of Engineering majors only.

Course Learning Outcomes (CLOs)

1. **CLO 1: Analyze and Manipulate Signals:** Characterize discrete-time signals and systems in both the time and frequency domains using the DTFT and z-transform.
2. **CLO 2: Evaluate Conversion Effects:** Mathematically model continuous-to-discrete (C/D) and discrete-to-continuous (D/C) conversions, mitigating effects like aliasing and quantization.
3. **CLO 3: Design Digital Filters:** Design FIR and IIR filters to meet specific magnitude criteria while evaluating critical properties like linear phase, group delay, and stability.
4. **CLO 4: Implement Multi-rate Systems:** Apply decimation, interpolation, and polyphase decomposition to design efficient sample-rate conversion systems.
5. **CLO 5: Perform Spectral Analysis:** Compute and interpret the Discrete Fourier Transform (DFT), Short-Time Fourier Transform (STFT), and Power Spectral Density (PSD) for both deterministic and random signals.
6. **CLO 6: Classify Audio Signals using ML:** Design and implement end-to-end machine learning pipelines (using Scikit-Learn and PyTorch) to extract time-frequency features from audio data (e.g., UrbanSound8K) and classify them using statistical and deep learning models.

Linking CLOs to Engineering Applications

Telecommunications: Software-defined radios (SDR) and digital transceivers. Up/down conversion relies heavily on multi-rate processing, while polyphase filters ensure computational efficiency at the edge (CLO 3, CLO 4, CLO 5).

Radar and Sonar Systems: Correlation, random signals, and PSD (CLO 5) are used for target tracking and signal recovery in noisy environments.

Embedded Edge AI: Designing efficient FIR/IIR filters (CLO 3) and migrating trained PyTorch audio classifiers to resource-constrained microcontrollers (CLO 6).

Hardware Accelerators: Understanding the DFT and polyphase architectures (CLO 4, CLO 5) to design custom FPGAs or ASICs for real-time DSP.

Acoustics & Environmental Monitoring: Using STFT and machine learning to classify city sounds and noise levels (CLO 5, CLO 6).

Biomedical Engineering: Filtering out 60 Hz powerline noise from ECGs (CLO 3) and classifying arrhythmias using random signal analysis and machine learning. Wearable health monitors. (CLO 5, CLO 6).

Topics Covered

- **Discrete-Time Signals and Systems (review)**
 - Linear and time-invariant systems
 - Convolution
 - Discrete-time Fourier transform
- **Sampling of Continuous-Time Signals**
 - CT to DT and DT to CT conversion
 - Sampling theorem
 - A/D and D/A conversion
- **The z-Transform (review)**
 - z-Transform properties
 - Pole-zero mapping
 - Frequency response of LTI systems
- **Filter Design**
 - FIR vs. IIR filters
 - Windowing methods
 - Bilinear transformation for IIR design
- **Filter Properties**
 - Time and frequency domain responses
 - Linear phase and generalized linear phase
 - Group delay
- **Multi-rate DSP**
 - Downsampling (decimation)
 - Upsampling (interpolation)
 - Frequency domain effects of rate changes
- **Polyphase Structures**
 - Polyphase decomposition
 - Multirate filterbanks
- **Random Signals**
 - Autocorrelation, cross-correlation
 - LTI system response to wide-sense stationary random inputs
- **Spectral Estimation**
 - Power Spectral Density (PSD)
 - Periodograms, Welch's method
- **Time-Frequency Analysis**
 - The Discrete Fourier Transform (DFT)
 - Fast convolution
 - Short-Time Fourier Transform (STFT)
- **Feature Engineering for Audio**
 - Mel-spectrograms and Mel-Frequency Cepstral Coefficients (MFCC)
 - Intro to Scikit-Learn
- **Traditional ML Classifiers**
 - Training SVMs, Decision Trees, and Random Forests on audio features

- Dimensionality reduction (PCA)
- **Introduction to PyTorch**
 - Tensors
 - Automatic differentiation
 - Multilayer Perceptrons (MLPs) for signal classification
- **Deep Learning for Audio**
 - Treating STFT spectrograms as images
 - Convolutional Neural Networks (CNNs) in PyTorch
 - Overfitting, dropout, and validation

Graduate versus Undergraduate Work

- **Homework**
 - Additional/harder problems for ECEN 5632
- **Quizzes, Exams**
 - Different questions and/or more strict grading criteria for ECEN 5632
- **Projects**
 - More extensive and/or more strict grading criteria for ECEN 5632

Homework/Projects

- Homework problems and projects will involve Python programming in Jupyter notebooks, analytic calculations, proofs, etc. The questions must be carefully and clearly answered to receive full credit. Complete sentences that establish a clear logical progression are highly recommended.
 - Students are encouraged to work together on homework problems and projects to **develop an understanding** of the material. However, submissions are individual and each student must:
 - Generate and turn in his/her own individual solutions that reflect his/her own individual level of understanding (this includes computer programs), and
 - Cite any external resources, including AI resources and prompts, used to answer a homework question (outside of the textbooks and lecture notes).
- In addition, students may not (unless explicitly directed otherwise):
- Copy homework solutions (including computer programs) from another student or from any other source including AI, or
 - Consult homework solutions from any resource, including online and AI, pertaining directly to the questions in the homework.
- Violation of these policies will be considered violations of the Honor Code. Questions about these policies must be raised before handing in a homework assignment.

Exams

Section 001 (in class)

- There will be two midterm exams and one final exam.
- Midterm Exam 1: Thu, Sep 24, in class
- Midterm Exam 2: Thu, Nov 5, in class
- Final Exam: Mon, Dec 7, 1:30-4:00 pm (in regular classroom)

Section 002 (on Zoom, asynchronous)

Students are responsible to arrange for proctoring their exams or take the exams in class at the times specified for Section 001.

- There will be two midterm exams and one final exam.
- Midterm Exam 1: Within a week of Sep 24, arrange for proctor
- Midterm Exam 2: Within a week of Nov 5, arrange for proctor
- Final Exam: Within two days of Dec 7, arrange for proctor

Course Requirements and Grading (tentative)

- Class participation and attendance will count for 10% of the course grade.
- The percentages of different course components toward the final grade differ between ECEN 4632 and ECEN 5632, with details given below:

Category	Weight for 4632	Weight for 5632
Homework	30%	15%
Quizzes	10%	10%
Midterm Exams	20%	30%
Final Exam	15%	20%
Projects	20%	20%
Exit Tickets	5%	5%

- Exit tickets will be brief assessments of student understanding of the material taught in class. They will be administered approximately weekly on Canvas in the form of an essay question.